SUMMARY MY APPROACH

Model

Deep Neural Network

- 3 hidden layers
- dim: 5(input) -> 128 -> 64 -> 32 -> 32 -> 1(output)
- batch size : 256
- learning rate = 0.001

Optimization

Bayesian optimization

- surrogate model : Gaussian Process
- acquisition function: EI with trade off between exploration and exploration(=0.01)

Result

On average, individuals who invest more time in reading a book, solving examples, and discussing with friends than in taking a rest and getting sleep tend to have higher scores.

Therefore, To get the highest score,
you need to invest a lot of time in study-related activities, and you need adequate rest and sleep.

This is query summary

```
df.describe()
```

	A	B	C	D	E
count	10.000000	10.000000	10.000000	10.000000	10.000000
mean	0.713391	0.641421	0.175428	0.826233	0.417255
std	0.136490	0.074780	0.080804	0.108201	0.094019
min	0.440665	0.500465	0.053039	0.601016	0.294742
25%	0.632690	0.608717	0.106356	0.784075	0.347754
50%	0.758716	0.651032	0.176778	0.826086	0.386314
75%	0.787922	0.695415	0.247472	0.885835	0.491471
max	0.874240	0.729726	0.288318	1.000000	0.554257

```
# Load library
#!pip install bayesian-optimization
from bayes_opt import BayesianOptimization
from bayes_opt import UtilityFunction
from sklearn.model_selection import train_test_split

import torch
import torch.nn as nn
import torch.optim as optim

import numpy as np
import pandas as pd
import seaborn as sns
from tqdm import tqdm
import matplotlib.pyplot as plt
```

Step1: Data Preparation

Get Offline data from csv file and Split into train and validation set

This data consists of pairs of decision variables and their corresponding exam scores

Get Offline data from csv file

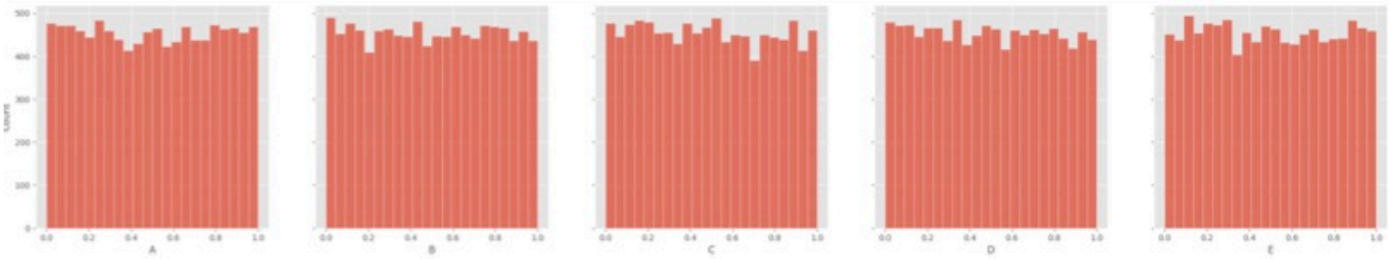
Decision variables

- A: Reading a Book
- B: Solving Examples
- C: Taking a Rest
- D: Discussing with Friends
- E: Getting Sleep

```
# Load the offline dataset containing pairs of decision variables and exam scores
data = pd.read_csv('/content/log.csv')
data.describe()
```

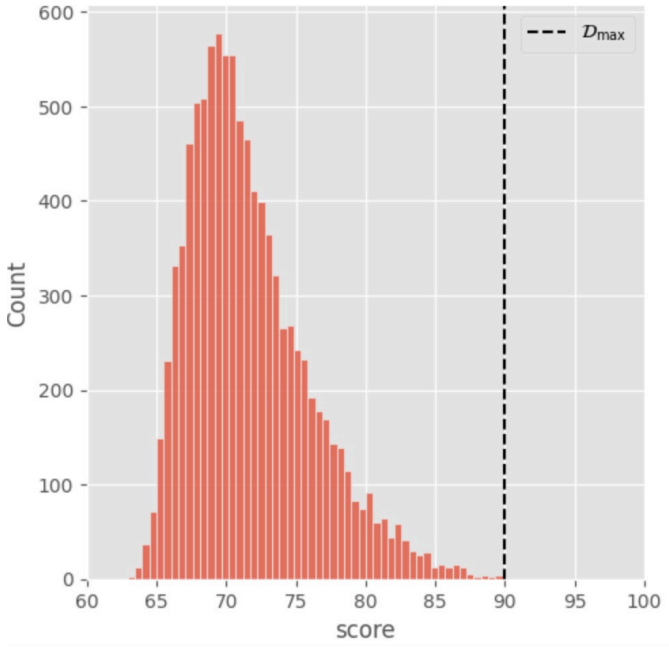
	A	B	C	D	E	score
count	9992.000000	9992.000000	9992.000000	9992.000000	9992.000000	9992.000000
mean	0.498985	0.498149	0.494426	0.495300	0.498725	71.645244
std	0.291844	0.289512	0.289011	0.289082	0.289978	4.333470
min	0.000064	0.000010	0.000397	0.000126	0.000233	62.944380
25%	0.244372	0.247665	0.241093	0.242532	0.245855	68.460217
50%	0.499789	0.498503	0.491682	0.494536	0.497319	70.780107
75%	0.754511	0.749839	0.747202	0.743745	0.750852	74.079871
max	0.999860	0.999969	0.999882	0.999996	0.999987	89.830920

```
# Visualize Data Distribution
plt.style.use("ggplot")
fig, axes = plt.subplots(1, 5, figsize=(30, 5), sharey=True)
for i in range(5):
    sns.histplot(data.iloc[:, i], ax=axes[i])
plt.show()
```

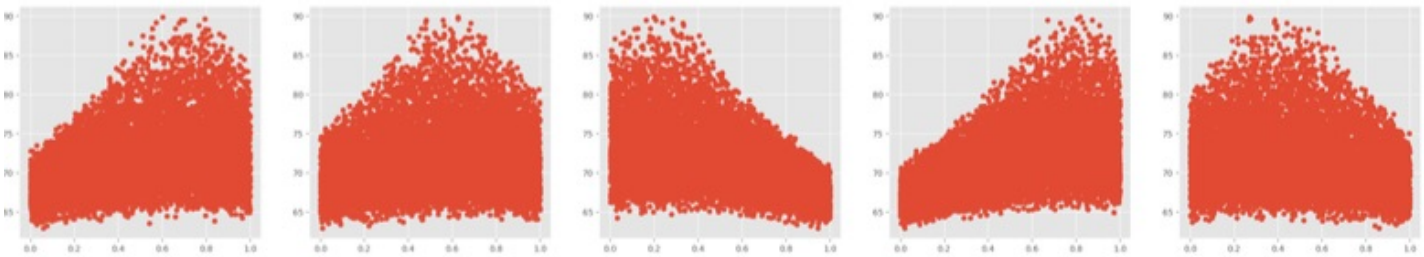


```
# Visualize Data Distribution
plt.style.use("ggplot")
plt.figure(dpi=120)
sns.displot(data['score'])
plt.xlim(60, 100)
plt.axvline(90, linestyle="--", color="black", label="$\mathcal{D}_{\max}$")
plt.legend()
```

```
<matplotlib.legend.Legend at 0x7cdbec2bdb40>
<Figure size 768x576 with 0 Axes>
```




```python
# Visualize Data Distribution
plt.style.use("ggplot")
plt.figure(figsize=(30, 5))
for i in range(data.shape[1]-1):
    plt.subplot(1, 5, i+1)
    plt.scatter(data.iloc[:, i], data["score"])
```



Visualize Data Distribution

1. all of A, B, C, D, E are similar distribution
2. skewness of score is positive
3. "A", "D" seems like weak positive correlation of score

"B" seems like weak negative correlation of score

```python
data[data["score"] == max(data["score"])]
```

	A	B	C	D	E	score
8446	0.602218	0.626019	0.199243	0.81822	0.270729	89.83092

Maximum score in the data

The person who scored the highest spent more time reading a book (A), solving examples (B), and discussing with friends (D) than taking a rest (C) and getting sleep (E)

Split into Train and Validation set

```python
# Split into Train and Validation set
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

df = torch.from_numpy(data.values).float().to(device)
x, y = df[:, :5], df[:, -1:]

num_points = x.shape[0]
train_x, valid_x = x[:int(num_points * 0.8)], x[int(num_points * 0.8):]
train_y, valid_y = y[:int(num_points * 0.8)], y[int(num_points * 0.8):]
```

Step2: Model Training

Choose model and Train the model using the training data

Simple DNN using 3 hidden layers

Ultimately, I configured a simple DNN by experimenting with different hyperparameter values.

I varied the hidden dimension {16, 32, 64, 128, 256} for each layer and batch size {128, 256, 512}, adjusted the number of layers {2, 3, 4}, and added dropout to prevent overfitting.

```
class BaseMLP(nn.Module):
    def __init__(self, input_dim, output_dim, hidden_dim, dropout):
        super(BaseMLP, self).__init__()

        self.model = nn.Sequential(nn.Linear(input_dim, hidden_dim),
                                   nn.ReLU(),
                                   nn.Dropout(dropout),
                                   nn.Linear(hidden_dim, int(hidden_dim/2)),
                                   nn.ReLU(),
                                   nn.Dropout(dropout),
                                   nn.Linear(int(hidden_dim/2), int(hidden_dim/4)),
                                   nn.ReLU(),
                                   nn.Linear(int(hidden_dim/4), int(hidden_dim/4)),
                                   nn.ReLU(),
                                   nn.Linear(int(hidden_dim/4), output_dim))

    def forward(self, x):
        return self.model(x)
```

```
# Hyperparameters for training neural network
batch_size = 256
training_step = 10000
validation_interval = 1000
lr = 1e-3
```

```
# Define optimizer and loss function
model = BaseMLP(input_dim=5, output_dim=1, hidden_dim=128, dropout=0.2)
optimizer = optim.Adam(model.parameters(), lr=lr)
criterion = nn.MSELoss()
```

Training the model and Set the objective function

```
# Train a neural network
model = model.to(device)

train_losses = []
valid_losses = []
for step in range(training_step):
    idx = torch.randint(0, len(train_x), size=(batch_size, ))

    batch_train_x, batch_train_y = train_x[idx], train_y[idx]
    batch_pred_y = model(batch_train_x)

    optimizer.zero_grad()

    train_loss = criterion(batch_train_y, batch_pred_y)
    train_loss.backward()
    optimizer.step()

    train_losses.append(train_loss.item())

    if (step+1) % validation_interval == 0:
        valid_loss = criterion(model(valid_x), valid_y)
        valid_losses.append(valid_loss.item())

        print(f"Step: {step+1}/{training_step}\tTrain Loss: {train_losses[-1]:.2f}\tValid Loss: {valid_losses[-1]:
```

```
Step: 1000/10000        Train Loss: 42.84       Valid Loss: 42.16
Step: 2000/10000        Train Loss: 22.96       Valid Loss: 22.38
Step: 3000/10000        Train Loss: 6.73        Valid Loss: 5.99
Step: 4000/10000        Train Loss: 3.26        Valid Loss: 3.46
Step: 5000/10000        Train Loss: 2.20        Valid Loss: 2.36
Step: 6000/10000        Train Loss: 1.73        Valid Loss: 1.88
Step: 7000/10000        Train Loss: 1.80        Valid Loss: 1.72
Step: 8000/10000        Train Loss: 1.48        Valid Loss: 1.21
Step: 9000/10000        Train Loss: 1.27        Valid Loss: 1.22
Step: 10000/10000       Train Loss: 1.23        Valid Loss: 1.14
```

```python
# Objective function to maximize the output of DNN
def bo_objective(A, B, C, D, E):
    device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
    dtype = torch.float32  # setting the data type

    # Convert input x to tensor and add batch dimension
    x = torch.tensor([A, B, C, D, E], device=device, dtype=dtype).unsqueeze(0)

    return model(x).item()   # score: model(x)
```

Step3: Optimization Algorithm

$x^* = argmax_x f(x)$

subject to $0 \leq x \leq 1$

- $f(x)$: black-box function

Bayesian Optimization

$P(Model|data) \propto P(data|Model) * P(Model) \Leftrightarrow Posterior \propto Likelihood * Prior$

Surrogate model

> Probabilistic model to estimate the distribution on unknown target values f = f(x) corresponding x

1. Data

$x \rightarrow f(x) \rightarrow y$

- $x_1, \ldots, x_n$
- $y_i = f_i(x) + \epsilon_i, i = (1, \ldots, n)$

2. Prior of the function values

$$\begin{bmatrix} f_1 \\ \vdots \\ f_n \end{bmatrix} \sim GP\left(\begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} k(x_1, x_1) & \cdots & k(x_1, x_n) \\ \vdots & \ddots & \vdots \\ k(x_n, x_1) & \cdots & k(x_n, x_n) \end{bmatrix}\right)$$

- $m(x) = 0$
- $k(x, x') = \gamma \exp(-\frac{1}{2}(x - x')^T diag(\lambda)^- 2(x - x'))$

3. Likelihood based on Gaussian noise

$(y_1, \ldots, y_n)|(f_1, \ldots, f_n) \sim N((f_1, \ldots, f_n)^T, \sigma_\epsilon^2 I)$

- To find parameter, we maximize the log-likelihood of the training data $D = (x_i, y_i)$

4. Joint distribution based on Bayes' rule

$$\begin{bmatrix} (y_1, .., y_n) \\ f \end{bmatrix} \sim GP(0, \begin{bmatrix} K + \sigma_\epsilon^2 I & k \\ k^T & k(x, x) \end{bmatrix})$$

5. Conditionalization: Posterior distribution f given data

$f|D \sim N(\mu(x|D), \sigma^2(x|D))$

- $\mu(x|D) = k^T(K + \sigma_\epsilon^2 I)^{-1} y_{1:n}$
- $\sigma^2(x|D) = k(x, x) - k^T(K + \sigma_\epsilon^2 I)^{-1} k$

Acquisitic function

> Function that suggests candidates to look into to find the optimal value based on predicted mean and variance

$x^{(n+1)} = argmax_x EI(x)$

- Exploration : uncertainty is great
- Exploitation : prediction is high

1. UCB (Upper Confidence Bound)

```
next input is maximum of Upper bound on uncertainty resulting from GP regression
```

$UCB(X) = \mu(x) + \kappa\sigma(x)$

---

2. PI (Probability Improvement)

```
probability of improving our current best estimate
```

$PI(x) = p(f(x) \geq f(x^+)) = \Phi(\frac{\mu(x;\theta,D) - f(x^+) - \xi}{\sigma(x;\theta,D)})$

- $f_{max}(x) = f(x^+)$
- ξ : trade off between Exploration and Exploitation

  $\xi \uparrow, Exploration \uparrow$: new input point move to great uncetainty

---

3. EI (Expected Improvement)

```
how much better improving our current best estimate
```

$EI(x) = (\mu(x) - f(x^+) - \xi)\Phi(Z) + \sigma(x)\phi(Z)$

- Z = $\frac{\mu(x;\theta,D) - f(x^+) - \xi}{\sigma(x;\theta,D)} * I(\sigma(x > 0))$

My Bayesian Optimization

- surrogate model

  In bayesian-optimization package, Surrogate model is fixed to Gaussian Process

- acquisitic function

  I choose Expected Improvemnt

bayesian-optimization package

1. BayesianOptimization

- f : function to be optimized (simple DNN)
- pbounds : parameters with their corresponding bounds (range of X)

2. optimizer.maximize

- init_points : # of steps of random exploration
- n_iter : # of steps of bayesian optimization
- acq : acquisition function
- xi : controls the relative intensity between exploration and exploration

```python
# Find queries that maximizes the trained neural network
candidates = []
num_trials = 10 # SHOULD BE FIXED

# Search space for each input dimension
pbounds={'A': (0, 1), 'B': (0, 1), 'C': (0, 1), 'D': (0, 1), 'E': (0, 1)}

for trial in range(num_trials):

  # Initialize Bayesian Optimization
  optimizer = BayesianOptimization(
          f=bo_objective,  # bo_objective is black-box function (trained DNN)
          pbounds=pbounds)

  # Find queries that maximizes the trained simple DNN
  acq = UtilityFunction(kind="ei", xi=0.01)
  optimizer.maximize(init_points=5, n_iter=100, acquisition_function=acq)

  candidates.append(optimizer.max['params'])
```

↳ this result in last page

Step 4: Reporting Results

```python
# Save data into query.csv file
df = pd.DataFrame(candidates, columns=["A", "B", "C", "D", "E"])
df.to_csv("/content/query.csv", index=False)
df.head()
```

	A	B	C	D	E
0	0.740807	0.619585	0.241691	0.601016	0.348864
1	0.791495	0.555704	0.132013	0.850394	0.543378
2	0.776625	0.729726	0.088565	0.778261	0.487510
3	0.581368	0.606930	0.186078	0.752470	0.373789
4	0.874240	0.693442	0.097804	0.894010	0.347384

```python
df.describe()
```

	A	B	C	D	E
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std	0.136490	0.074780	0.080804	0.108201	0.094019
min	0.440665	0.500465	0.053039	0.601016	0.294742
25%	0.632690	0.608717	0.106356	0.784075	0.347754
50%	0.758716	0.651032	0.176778	0.826086	0.386314
75%	0.787922	0.695415	0.247472	0.885835	0.491471
max	0.874240	0.729726	0.288318	1.000000	0.554257

Result

On average, individuals who invest more time in reading a book, solving examples, and discussing with friends than in taking a rest and getting sleep tend to have higher scores.

Therefore, To get the highest score, you need to invest a lot of time in study-related activities, and you need adequate rest and sleep.

iter	target	A	B	C	D	E
1	69.33	0.4818	0.4827	0.9712	0.9138	0.1116
2	70.85	0.07939	0.5228	0.2476	0.3053	0.3911
3	76.07	0.7922	0.8767	0.1251	0.4366	0.1291
4	70.01	0.4881	0.3713	0.921	0.9363	0.7127
5	70.12	0.4346	0.7323	0.7715	0.329	0.5585
6	72.97	0.879	0.9329	0.008281	0.4228	0.07284
7	71.1	0.3991	0.6792	0.1382	0.09532	0.4914
8	76.35	0.7881	0.8677	0.1705	0.471	0.1219
9	75.46	0.7081	0.9364	0.2043	0.4719	0.1431
10	78.49	0.739	0.7578	0.1731	0.4439	0.1473
11	79.54	0.8538	0.9432	0.2563	0.4922	0.1307
12	76.37	0.904	0.9248	0.2974	0.4959	0.1305
13	74.48	0.8598	0.9739	0.2412	0.5146	0.1082
14	75.69	0.7734	0.8964	0.3059	0.411	0.171
15	67.05	0.7455	0.1921	0.3994	0.0825	0.9546
16	77.03	0.7778	0.8631	0.2929	0.4249	0.205
17	78.13	0.7766	0.7971	0.2128	0.439	0.1695
18	75.63	0.847	0.9181	0.2604	0.4781	0.1504
19	68.51	0.01929	0.3957	0.1805	0.3036	0.5312
20	75.33	0.8754	0.9965	0.3224	0.5162	0.166
21	78.07	0.6989	0.791	0.1371	0.4663	0.2089
22	78.45	0.7347	0.7934	0.1712	0.4517	0.1779
23	69.94	0.4231	0.5583	0.3716	0.2194	0.8228
24	70.6	0.8699	0.3217	0.7532	0.4664	0.6342
25	75.81	0.7516	0.8022	0.1775	0.4268	0.1376
26	81.17	0.7422	0.7558	0.1916	0.4596	0.1869
27	81.21	0.676	0.7342	0.2436	0.4597	0.2823
28	80.93	0.8992	0.7746	0.2111	0.7023	0.6191
29	71.28	0.2106	0.988	0.091	0.3779	0.684
30	70.45	0.05409	0.4205	0.4778	0.9556	0.7093
31	80.41	0.81	0.7058	0.2457	0.4779	0.2614
32	79.7	0.7322	0.7374	0.2256	0.4631	0.2415
33	68.74	0.6676	0.4933	0.814	0.1985	0.3591
34	80.5	0.7814	0.7325	0.1979	0.4747	0.2031
35	80.66	0.7334	0.7455	0.2056	0.4965	0.1837
36	69.59	0.5392	0.5308	0.7077	0.109	0.2789
37	68.98	0.4227	0.7537	0.8696	0.4862	0.1487
38	80.44	0.6548	0.7438	0.2245	0.4632	0.2425
39	80.11	0.669	0.7197	0.2125	0.491	0.2085
40	66.2	0.1901	0.3162	0.5206	0.02033	0.4037
41	80.95	0.6688	0.7763	0.2535	0.4837	0.2857
42	70.54	0.2376	0.6281	0.2463	0.1063	0.2537
43	69.29	0.3316	0.3102	0.33	0.2221	0.09953
44	78.29	0.7527	0.7172	0.2395	0.4528	0.1766
45	74.43	0.5142	0.7502	0.7157	0.5843	0.32
46	77.4	0.568	0.9909	0.1309	0.847	0.5138
47	80.41	0.6681	0.7694	0.2198	0.4444	0.2984
48	80.95	0.7215	0.7461	0.2718	0.476	0.3236
49	66.12	0.2442	0.9154	0.7806	0.02494	0.9996
50	66.76	0.2364	0.009613	0.6473	0.2562	0.6954
51	79.16	0.6551	0.7398	0.2843	0.4614	0.3204
52	80.14	0.7813	0.7545	0.2378	0.5104	0.3066
53	80.11	0.7911	0.7513	0.2692	0.4369	0.3122
54	73.3	0.5196	0.193	0.5805	0.9639	0.393
55	81.73	0.7737	0.6898	0.2408	0.4716	0.3334
56	83.61	0.7881	0.6872	0.2897	0.4959	0.3278
57	83.33	0.7094	0.7538	0.2874	0.4812	0.3403
58	67.31	0.5304	0.7478	0.7744	0.1254	0.8075
59	71.75	0.1438	0.9552	0.009982	0.6625	0.1734
60	76.05	0.6704	0.6639	0.2589	0.4595	0.1062
61	80.46	0.727	0.7549	0.3139	0.5023	0.3719
62	71.94	0.9857	0.005512	0.4383	0.4996	0.5144
63	71.57	0.5875	0.6872	0.6458	0.3216	0.00156
64	82.22	0.8262	0.6794	0.2747	0.4927	0.3389
65	83.69	0.7793	0.6513	0.2777	0.5177	0.32
66	69.59	0.2017	0.7489	0.7667	0.996	0.03904
67	78.7	0.7811	0.646	0.2988	0.4788	0.3245
68	70.24	0.1712	0.3402	0.7225	0.8186	0.4083
69	84.16	0.7867	0.6814	0.2694	0.5311	0.3241
70	66.82	0.2676	0.1866	0.9005	0.277	0.5742
71	80.23	0.6978	0.7996	0.2981	0.4736	0.3511
72	83.09	0.7592	0.6726	0.273	0.5281	0.3491
73	83.53	0.7715	0.6766	0.2995	0.5438	0.3005
74	68.16	0.1141	0.2035	0.2616	0.3693	0.849
75	85.3	0.7716	0.6449	0.2472	0.5522	0.2953
76	84.64	0.7951	0.624	0.257	0.5703	0.3037
77	76.0	0.2401	0.8183	0.193	0.9909	0.2842
78	84.72	0.7924	0.6263	0.2097	0.5523	0.3074
79	84.74	0.7507	0.6111	0.2298	0.5741	0.3094
80	70.38	0.1946	0.5072	0.567	0.4991	0.8313
81	67.47	0.5501	0.9407	0.9669	0.7511	0.9173
82	84.44	0.7754	0.6109	0.2245	0.5714	0.2613
83	83.17	0.7708	0.6461	0.2125	0.5983	0.2898
84	67.06	0.8242	0.005334	0.05581	0.1332	0.8327
85	68.18	0.3118	0.06173	0.235	0.1799	0.06396
86	68.79	0.08546	0.269	0.4157	0.3343	0.5787
87	82.2	0.7757	0.6154	0.2398	0.553	0.2964
88	85.53	0.7323	0.6335	0.2636	0.5937	0.3033
89	83.49	0.7193	0.5937	0.2355	0.6103	0.2784
90	86.35	0.7408	0.6196	0.2417	0.601	0.3489
91	82.96	0.767	0.6205	0.1945	0.5797	0.3707
92	82.29	0.7554	0.601	0.2701	0.6292	0.3395
93	81.5	0.7084	0.6283	0.2338	0.5867	0.3399
94	81.14	0.7583	0.6389	0.2633	0.591	0.226
95	71.91	0.565	0.4692	0.7896	0.4059	0.3568
96	70.22	0.9676	0.2746	0.07895	0.05583	0.879
97	68.11	0.006711	0.7949	0.8728	0.6124	0.5089
98	82.01	0.8393	0.62	0.1993	0.6075	0.2404
99	67.05	0.1237	0.7746	0.5179	0.1076	0.6018
100	75.74	0.3142	0.4941	0.2387	0.9516	0.6884
101	80.13	0.7101	0.7301	0.213	0.4682	0.1793
102	78.57	0.8201	0.6076	0.3085	0.4619	0.3118
103	84.1	0.762	0.6347	0.2406	0.591	0.3384
104	77.25	0.8244	0.2105	0.0235	0.6421	0.2292
105	66.99	0.9894	0.797	0.6343	0.01521	0.6804

iter	target	A	B	C	D	E
1	80.66	0.5704	0.4845	0.2043	0.4328	0.308
2	68.33	0.006659	0.9514	0.02025	0.1952	0.05393
3	69.71	0.474	0.9276	0.757	0.3421	0.2861
4	73.88	0.9313	0.8168	0.5693	0.5357	0.6117
5	67.86	0.02157	0.8541	0.7794	0.7765	0.6616
6	71.34	0.7135	0.4587	0.3048	0.1102	0.5203
7	69.57	0.6071	0.09936	0.73	0.473	0.7132
8	74.68	0.7999	0.9571	0.6704	0.7987	0.5637
9	74.52	0.4777	0.395	0.6352	0.7439	0.6967
10	81.97	0.5547	0.4825	0.214	0.4772	0.2977
11	80.91	0.5452	0.4648	0.2037	0.5713	0.272
12	69.02	0.7509	0.6114	0.7962	0.2419	0.2624
13	80.89	0.4082	0.5082	0.2691	0.5	0.3086
14	83.89	0.6073	0.5288	0.2671	0.5159	0.2595
15	73.15	0.9131	0.3406	0.3771	0.4538	0.7738
16	70.65	0.2758	0.605	0.6997	0.7806	0.06725
17	83.72	0.5762	0.5877	0.21	0.5298	0.2383
18	85.51	0.6612	0.6181	0.2444	0.5598	0.3037
19	72.01	0.4835	0.4534	0.7672	0.7481	0.3385
20	84.03	0.7734	0.6775	0.2236	0.5947	0.231
21	71.86	0.8665	0.585	0.4288	0.2914	0.1452
22	65.96	0.6886	0.8957	0.9552	0.05972	0.3837
23	83.61	0.7869	0.432	0.191	0.7366	0.3512
24	82.22	0.7565	0.6313	0.186	0.6891	0.3987
25	77.99	1.0	0.4174	0.2138	0.8335	0.1986
26	86.11	0.6655	0.6217	0.1911	0.5426	0.2732
27	86.05	0.7461	0.5609	0.1551	0.5662	0.2854
28	83.38	0.7564	0.6073	0.0	0.5473	0.2547
29	76.02	0.8687	0.18	0.0	0.8855	0.5538
30	67.1	0.0	0.0	0.0	1.0	1.0
31	65.29	0.8599	0.1157	0.8956	0.1545	0.08817
32	75.19	0.4691	0.4038	0.128	0.6053	0.8808
33	72.36	1.0	1.0	0.0	1.0	1.0
34	68.86	1.0	1.0	0.0	0.0	1.0
35	85.77	0.618	0.6027	0.2055	0.5517	0.209
36	79.65	0.4587	0.3591	0.3995	0.8123	0.2395
37	72.7	0.8455	0.2765	0.5226	0.4321	0.4179
38	82.81	0.6947	0.6011	0.1735	0.5869	0.1966
39	67.8	0.2496	0.878	0.8754	0.2608	0.3942
40	84.95	0.8043	0.5755	0.1416	0.5289	0.3555
41	68.15	0.09641	0.1295	0.8857	0.95	0.5152

42	81.3	0.6389	0.6885	0.253	0.5244	0.1876
43	69.79	0.05773	0.8385	0.3904	0.424	0.8723
44	85.79	0.7435	0.5633	0.2236	0.5604	0.3116
45	82.53	0.9789	0.5	0.04047	0.7062	0.3937
46	72.32	0.329	0.1429	0.2849	0.6579	0.7138
47	71.24	0.01533	0.3749	0.2733	0.6662	0.1489
48	78.4	0.2764	0.5297	0.4378	0.9808	0.3123
49	67.76	0.8599	0.9213	0.4017	0.1013	0.9545
50	81.58	0.9392	0.4009	0.01372	0.673	0.3983
51	80.15	0.936	0.7006	0.1106	0.6016	0.3034
52	77.39	0.8088	0.2023	0.313	0.8408	0.187
53	82.37	0.7202	0.5938	0.148	0.5569	0.3234
54	83.84	0.827	0.518	0.1622	0.5494	0.279
55	69.62	0.2855	0.2232	0.8127	0.6961	0.2115
56	69.11	0.008903	0.3835	0.3995	0.6573	0.06754
57	71.48	0.915	0.7357	0.1555	0.2185	0.8734
58	83.71	0.6812	0.5904	0.2228	0.5355	0.2501
59	77.82	0.4551	0.3686	0.438	0.8288	0.22
60	66.55	0.404	0.7523	0.7816	0.8224	0.1195
61	82.66	0.7087	0.549	0.1861	0.6279	0.3093
62	85.56	0.6258	0.6603	0.1896	0.5851	0.2529
63	85.43	0.6787	0.4885	0.2947	0.5999	0.368
64	85.66	0.7307	0.4704	0.19	0.5475	0.3211
65	85.73	0.7485	0.634	0.3221	0.6059	0.3846
66	80.48	0.4504	0.4806	0.2766	0.5021	0.3142
67	71.15	0.2946	0.04015	0.1325	0.5485	0.3385
68	67.12	0.3601	0.01758	0.716	0.258	0.7339
69	83.03	0.6743	0.597	0.3297	0.6886	0.36
70	69.6	0.233	0.4127	0.8686	0.928	0.03003
71	85.5	0.7728	0.7198	0.2589	0.5647	0.3415
72	84.18	0.7947	0.5645	0.2763	0.5346	0.4042
73	71.44	0.6501	0.8391	0.8754	0.838	0.3314
74	69.87	0.06729	0.2154	0.05576	0.6734	0.5163
75	84.6	0.8483	0.4899	0.2355	0.7671	0.4955
76	81.69	0.7644	0.5461	0.07742	0.4991	0.2473
77	83.34	0.7437	0.7682	0.1337	0.5252	0.253
78	81.19	0.8411	0.6668	0.3016	0.6956	0.4665
79	84.93	0.7381	0.4006	0.255	0.7183	0.4678
80	72.42	0.7808	0.2677	0.2442	0.183	0.3454
81	84.9	0.7482	0.4422	0.2128	0.8582	0.4597
82	66.85	0.6063	0.4942	0.8735	0.949	0.5446
83	65.58	0.1603	0.9839	0.9373	0.3134	0.9599
84	84.37	0.8242	0.4042	0.1482	0.7847	0.4798
85	85.73	0.8782	0.5399	0.1158	0.8881	0.4638
86	85.7	0.682	0.5909	0.2634	0.5593	0.3205
87	87.04	0.7915	0.5557	0.132	0.8584	0.5434
88	68.33	0.2643	0.09279	0.8474	0.9016	0.07648
89	71.34	0.7143	0.9825	0.2643	0.2014	0.2967
90	81.23	0.6881	0.6322	0.1734	0.6177	0.201
91	81.65	0.8668	0.6968	0.07242	0.8699	0.514
92	85.09	0.834	0.5168	0.1696	0.9204	0.5857
93	82.58	0.7475	0.5265	0.07293	0.9185	0.5075
94	84.3	0.869	0.5308	0.1114	0.8043	0.5602
95	86.72	0.7888	0.5725	0.1976	0.8616	0.5118
96	71.45	0.2016	0.2235	0.3387	0.6642	0.2081
97	84.1	0.7251	0.5355	0.2009	0.8063	0.5785
98	85.84	0.911	0.511	0.2624	0.9708	0.4478
99	71.51	0.1269	0.6907	0.02615	0.8569	0.6835
100	81.26	1.0	0.4477	0.2016	0.9215	0.4696
101	82.55	0.8485	0.6041	0.2295	1.0	0.4491
102	85.04	0.8414	0.4466	0.3624	0.9482	0.4728
103	72.17	0.8632	0.54	0.4157	0.1623	0.616
104	80.07	0.9704	0.522	0.4255	1.0	0.4177
105	65.29	0.09174	0.2293	0.8501	0.8921	0.2267

iter	target	A	B	C	D	E
1	66.36	0.3134	0.8856	0.7137	0.005373	0.7857
2	79.2	0.6266	0.5568	0.4878	0.8085	0.721
3	67.62	0.07298	0.0917	0.4997	0.1321	0.4239
4	70.02	0.9094	0.1986	0.9216	0.9227	0.4281
5	80.9	0.8018	0.9405	0.366	0.75	0.4835
6	76.54	0.6799	0.6006	0.5107	0.8446	0.7787
7	68.75	0.4826	0.3041	0.9941	0.7793	0.4217
8	78.48	0.6438	0.4928	0.4558	0.8614	0.7307
9	83.7	0.633	0.6721	0.325	0.7437	0.5124
10	83.31	0.4821	0.5872	0.1621	0.7376	0.5462
11	73.76	0.6922	0.4267	0.4645	0.2524	0.3525
12	85.75	0.5053	0.7257	0.2472	0.9037	0.3848
13	82.87	0.6176	0.7666	0.05481	0.9893	0.1636
14	75.07	0.3193	0.9215	0.2077	1.0	0.4144
15	81.42	0.6735	0.7666	0.03388	0.958	0.1406
16	81.76	0.6124	0.5545	0.4494	0.8373	0.6557
17	84.68	0.5564	0.5797	0.2521	0.9591	0.3444
18	84.56	0.6147	0.7123	0.2071	0.943	0.4381
19	81.23	0.5593	0.7199	0.3367	0.8752	0.2968
20	82.47	0.6653	0.6495	0.2368	0.9216	0.3967
21	82.37	0.4647	0.6252	0.2037	0.9322	0.4304
22	68.83	0.2897	0.2681	0.5267	0.2588	0.6032
23	84.6	0.5409	0.7979	0.1647	0.798	0.4426
24	69.25	0.5735	0.1762	0.02781	0.03994	0.4837
25	75.69	0.5505	0.1263	0.07523	0.4512	0.4387
26	83.18	0.5316	0.7587	0.3032	0.8407	0.5433
27	80.27	0.5452	0.5013	0.1304	1.0	0.1377
28	84.27	0.5645	0.7164	0.289	0.8529	0.5291
29	81.18	0.7072	0.7492	0.023	0.6262	0.6633
30	78.11	0.7326	1.0	0.0	1.0	0.4076
31	84.93	0.5377	0.7387	0.1784	0.9244	0.3413
32	69.66	1.0	1.0	0.0	0.3402	1.0
33	66.48	0.0	0.0	0.0	1.0	1.0
34	72.38	1.0	0.0	0.0	1.0	0.0
35	82.38	0.6378	0.7326	0.07831	0.9817	0.1471
36	81.33	0.5246	0.7119	0.0	0.5646	0.3968
37	64.6	1.0	0.0	1.0	0.0	0.0
38	68.65	1.0	1.0	0.0	0.0	0.0
39	70.18	1.0	1.0	1.0	1.0	0.0
40	73.63	0.4266	0.9838	0.5501	0.8854	0.5775
41	83.97	0.5336	0.4756	0.3415	0.7695	0.4711
42	82.73	0.5382	0.7074	0.2793	1.0	0.3828
43	77.27	0.5917	0.562	0.4729	0.8309	0.7317
44	70.05	0.1704	0.9525	0.219	0.2445	0.4542
45	69.85	0.8275	0.3069	0.8887	0.626	0.7327
46	82.34	0.7172	0.4783	0.3125	0.7133	0.6004
47	84.41	0.5151	0.6765	0.2817	0.9987	0.3856
48	82.19	0.4524	0.5664	0.3746	0.9716	0.3556
49	85.73	0.5353	0.6959	0.2264	0.8437	0.3996
50	84.77	0.6636	0.7103	0.06044	0.8563	0.525
51	71.29	0.7251	0.2803	0.8701	0.9642	0.2531
52	75.98	0.344	0.7804	0.01976	0.804	0.08464
53	81.3	0.8433	0.6298	0.0	1.0	0.6701
54	84.38	0.5537	0.6784	0.2856	0.8811	0.5346
55	84.78	0.6248	0.546	0.0205	0.7834	0.4991
56	65.98	0.7522	0.3332	0.9714	0.07329	0.4791
57	79.88	0.6452	0.4964	0.0	0.7558	0.6705
58	65.88	0.3189	0.1287	0.9882	0.01853	0.6738
59	84.67	0.7351	0.6588	0.0	0.7152	0.4255
60	84.94	0.6323	0.6356	0.1152	0.6947	0.4828
61	83.28	0.6464	0.6657	0.0	0.9325	0.3634
62	71.9	1.0	0.0	0.0	1.0	1.0
63	83.15	1.0	0.766	0.0	0.7823	0.5272
64	76.56	0.7855	0.7089	0.0635	0.386	0.5792
65	78.77	0.9253	0.5828	0.002759	0.9861	0.8227
66	80.36	1.0	0.7769	0.2607	0.7986	0.6456
67	79.81	1.0	0.5017	0.0	0.6926	0.4429
68	66.45	0.0	1.0	1.0	1.0	0.0
69	76.23	1.0	1.0	0.0	0.6625	0.4003
70	68.62	0.0	0.0	0.0	0.4813	0.0
71	69.94	0.8747	0.4547	0.0281	0.6748	0.1075
72	77.77	0.8402	0.3035	0.609	0.6975	0.4543
73	81.59	0.6535	0.5046	0.4705	0.7605	0.5177
74	67.1	1.0	1.0	1.0	1.0	1.0
75	82.08	0.5671	0.486	0.1534	0.6382	0.4297
76	67.51	0.7318	0.9569	0.8538	0.6544	0.964
77	70.96	0.8807	0.2668	0.8114	0.5583	0.4705
78	74.27	0.9105	0.6257	0.6836	0.7783	0.001878
79	65.89	0.0	0.0	1.0	1.0	0.0
80	84.6	0.5275	0.5869	0.2559	0.9164	0.3622
81	67.73	1.0	0.0	0.0	0.0	1.0
82	86.86	0.7766	0.7297	0.08857	0.7783	0.4875
83	65.38	1.0	1.0	1.0	0.0	0.0

84	82.45	0.8097	0.7773	0.0999	0.8114	0.5406
85	81.97	0.7041	1.0	0.02155	0.9968	0.43
86	82.84	0.7744	0.6832	0.0979	0.7593	0.4652
87	71.54	0.03598	0.3885	0.6232	0.8089	0.3249
88	80.53	0.7623	0.8853	0.07183	0.6627	0.3953
89	82.15	0.6592	0.8621	0.07905	0.8249	0.3568
90	82.98	0.6979	0.7668	0.06022	0.8607	0.5471
91	72.61	0.7825	0.6193	0.7522	0.9373	0.01158
92	84.83	0.7045	0.7759	0.00839	0.8086	0.5827
93	83.85	0.65	0.8196	0.008285	0.7969	0.4836
94	83.82	0.7011	0.6634	0.04629	0.7083	0.5781
95	65.69	0.3222	0.4033	0.9506	0.1028	0.5228
96	64.92	0.04626	0.3056	0.09314	0.005613	0.7718
97	76.41	0.5008	1.0	0.09193	1.0	0.5665
98	81.41	0.6307	0.3799	0.2724	1.0	0.4974
99	83.57	0.6385	0.6929	0.04421	0.7262	0.4952
100	72.47	0.3791	0.2118	0.3	0.6114	0.005624
101	78.01	1.0	0.6149	0.0	0.7914	0.7563
102	84.73	0.7557	0.7615	0.04991	0.761	0.4871
103	83.38	0.6922	0.7432	0.1547	0.7813	0.5353
104	83.95	0.8205	0.6812	0.0	0.8314	0.4726
105	82.88	0.5978	0.7567	0.0564	0.7844	0.6858

iter	target	A	B	C	D	E
1	74.8	0.9668	0.9824	0.4766	0.9804	0.5852
2	67.04	0.2347	0.1917	0.001858	0.02409	0.8117
3	72.27	0.3403	0.05342	0.05084	0.6115	0.67
4	68.03	0.6045	0.03365	0.749	0.5011	0.8176
5	67.52	0.4997	0.4636	0.8473	0.178	0.8162
6	77.72	0.9901	0.9786	0.5197	0.9666	0.4881
7	75.79	1.0	0.984	0.5532	0.9645	0.4065
8	75.24	1.0	0.9232	0.534	0.9861	0.4976
9	78.69	0.8253	0.6362	0.3685	0.5743	0.1286
10	72.7	0.8335	0.5929	0.3039	0.1478	0.7629
11	65.83	0.7923	0.9874	0.8762	0.09532	0.8129
12	75.64	0.2324	0.7519	0.1849	0.5317	0.4264
13	71.91	0.4498	0.42	0.2836	0.5386	0.9562
14	82.18	0.7655	0.6604	0.3259	0.5884	0.164
15	81.62	0.714	0.6783	0.2927	0.5967	0.191
16	83.91	0.7929	0.6901	0.2842	0.6363	0.1903
17	85.68	0.7846	0.688	0.2596	0.6985	0.2427
18	84.44	0.7993	0.6137	0.2508	0.7447	0.234
19	84.9	0.8459	0.6872	0.2264	0.6976	0.3162
20	66.41	0.01913	0.6376	0.9276	0.8714	0.9035
21	84.15	0.8144	0.7511	0.2242	0.7981	0.2438
22	77.14	0.8598	0.3398	0.5349	0.852	0.1862
23	85.72	0.7911	0.6992	0.3319	0.729	0.296
24	84.84	0.7377	0.6856	0.2587	0.7662	0.345
25	83.82	0.6895	0.7123	0.3175	0.984	0.2477
26	81.43	0.6895	0.7061	0.3704	0.8188	0.2159
27	84.47	0.6004	0.6954	0.1114	1.0	0.3017
28	80.97	0.5516	0.8795	0.1895	1.0	0.3156
29	71.72	0.9621	0.838	0.7749	0.5185	0.3249
30	81.43	0.6257	0.5585	0.2306	1.0	0.3553
31	69.99	0.68	0.9417	0.1326	0.1764	0.6025
32	72.02	0.9203	0.934	0.5374	0.2857	0.1969
33	83.63	0.7049	0.6618	0.3906	0.7702	0.2216
34	70.89	0.7752	0.8354	0.8108	0.3858	0.2732
35	73.96	0.7709	0.6306	0.04751	0.333	0.7968
36	71.26	0.3188	0.3118	0.6607	0.5215	0.5455
37	79.17	0.5016	0.6563	0.1681	1.0	0.1439
38	84.89	0.7587	0.74	0.1367	1.0	0.3176
39	84.46	0.7719	0.6427	0.3316	0.6897	0.2873
40	81.62	0.728	0.717	0.0	0.9063	0.3894
41	83.62	0.8071	0.6528	0.004366	1.0	0.1674
42	68.42	0.02795	0.9405	0.2168	0.2024	0.2988
43	66.73	0.1373	0.4434	0.7519	0.2214	0.1618
44	78.02	0.7656	0.6233	0.3851	0.6298	0.145
45	83.13	0.725	0.6671	0.1771	1.0	0.238
46	79.0	0.9888	0.865	0.02261	0.9569	0.1654
47	70.85	0.2053	0.4315	0.1554	0.5359	0.9131
48	83.71	0.6847	0.6131	0.4514	0.865	0.3543
49	84.03	0.513	0.703	0.3342	0.9619	0.3881
50	79.34	0.4929	0.6311	0.512	1.0	0.2936
51	84.0	0.9973	0.538	0.0	1.0	0.2896
52	77.57	0.7364	0.6147	0.2403	0.4201	0.09362
53	79.99	1.0	0.435	0.0	1.0	0.08076
54	80.6	1.0	0.4386	0.0	1.0	0.5176
55	68.42	0.2843	0.9764	0.2696	0.1997	0.02743
56	83.0	0.9404	0.5931	0.0	0.8021	0.2452
57	70.06	0.2341	0.9894	0.2209	0.3568	0.3457
58	68.75	0.9144	0.9515	0.998	0.7627	0.408
59	83.57	0.9385	0.678	0.182	0.9377	0.3416
60	71.12	0.6883	0.7372	0.4994	0.4638	0.9387
61	82.48	0.7946	0.6754	0.2366	0.7611	0.2976
62	71.08	0.2762	0.1167	0.3511	0.7641	0.01002
63	85.14	0.7835	0.679	0.187	0.6806	0.3034
64	82.91	0.6776	0.5933	0.4395	0.8696	0.3277
65	84.21	0.6899	0.7434	0.326	0.7503	0.402
66	81.27	0.8452	0.8524	0.1073	0.5954	0.3577
67	82.01	0.6658	0.739	0.2794	1.0	0.3957
68	81.46	0.8983	0.6573	0.1654	0.6215	0.2435
69	86.05	0.5951	0.6551	0.2263	0.7838	0.3746
70	69.31	0.8552	0.4953	0.9812	0.6168	0.4662
71	83.79	0.8141	0.7725	0.2687	0.6718	0.3155
72	82.81	0.6459	0.636	0.3121	0.7775	0.3879
73	70.78	0.1288	0.3106	0.4413	0.5181	0.7258
74	85.28	0.5688	0.7091	0.1371	0.7548	0.3154
75	79.17	0.6245	0.7223	0.1833	0.7853	0.384
76	85.75	0.5414	0.6151	0.1875	0.7694	0.3092
77	85.41	0.5827	0.6547	0.2254	0.7597	0.3197
78	85.77	0.5273	0.6225	0.1967	0.8115	0.3837
79	82.02	0.5649	0.6497	0.05085	0.7385	0.2698
80	83.14	0.8777	0.6714	0.02133	1.0	0.328
81	83.02	0.4907	0.6715	0.1783	0.8481	0.3308
82	86.43	0.5814	0.6069	0.1861	0.7525	0.3738
83	81.44	0.7498	0.9112	0.124	0.934	0.2079
84	81.61	0.8112	0.7282	0.1232	0.789	0.103
85	84.25	0.8233	0.4612	0.0	0.9409	0.2899
86	69.87	0.2307	0.9158	0.7191	0.6199	0.5506
87	72.47	0.3884	0.3035	0.2453	0.3162	0.7271
88	85.86	0.7836	0.7195	0.4032	0.8022	0.373
89	84.7	0.5816	0.628	0.1672	0.7219	0.375
90	83.41	0.8364	0.3306	0.1241	0.7514	0.3162
91	66.93	0.03574	0.531	0.62	0.3142	0.1429
92	81.85	0.9415	0.3242	0.04406	0.8722	0.3065
93	82.93	0.9091	0.5058	0.1231	0.7124	0.4542
94	83.0	0.5981	0.6045	0.2007	0.8023	0.3586
95	82.0	0.5296	0.6251	0.2296	0.75	0.3923
96	69.19	0.2178	0.4539	0.01486	0.2869	0.0279
97	84.21	0.9292	0.5092	0.12	0.9191	0.277
98	69.24	0.5035	0.7862	0.9813	0.8665	0.1447
99	82.98	0.8061	0.679	0.3316	0.733	0.4301
100	83.63	0.8221	0.656	0.4328	0.8509	0.2986
101	85.22	0.6829	0.7526	0.0152	1.0	0.2529
102	85.06	0.8432	0.4565	0.0	0.7691	0.3497
103	85.17	0.5328	0.6251	0.1456	0.7748	0.354
104	70.4	0.7537	0.7245	0.9478	0.9607	0.5744
105	83.14	0.7511	0.4943	0.1337	0.6552	0.3204

iter	target	A	B	C	D	E
1	73.53	0.2711	0.3607	0.4229	0.5078	0.3184
2	65.45	0.1294	0.9787	0.7116	0.2953	0.9022
3	66.47	0.622	0.7938	0.6414	0.02624	0.798
4	75.51	0.4854	0.248	0.204	0.9	0.1498
5	70.34	0.9244	0.1662	0.7731	0.3493	0.5039
6	71.44	0.1191	0.0	0.04919	0.8587	0.0
7	77.49	0.5974	0.4536	0.3685	0.9127	0.03691
8	77.88	0.824	0.7026	0.2237	1.0	0.0
9	75.56	0.5229	0.8613	0.4026	1.0	0.0
10	77.61	1.0	0.4383	0.3429	1.0	0.0
11	77.54	0.947	0.5822	0.0	0.7253	0.0
12	79.08	0.9321	0.5599	0.0	1.0	0.0
13	85.88	1.0	0.5896	0.0	1.0	0.2927
14	83.37	1.0	0.6189	0.0	1.0	0.499
15	81.46	1.0	0.4426	0.0	1.0	0.3387
16	71.11	0.4194	0.02503	0.4198	0.5877	0.6065
17	71.08	0.9135	0.7767	0.2666	0.1492	0.7297

18	85.23	1.0	0.7325	0.0	1.0	0.3034
19	70.83	0.9901	0.1958	0.8189	0.7003	0.5136
20	83.4	0.9449	0.668	0.009246	0.9774	0.3024
21	76.08	1.0	1.0	0.0	1.0	0.6739
22	69.0	1.0	1.0	1.0	1.0	0.0
23	78.24	0.6491	0.2159	0.1318	0.7312	0.3189
24	83.92	0.9852	0.6733	0.03614	0.9522	0.2943
25	68.94	1.0	0.0	0.0	0.02245	0.0
26	65.91	0.0	1.0	0.0	0.0	0.0
27	73.09	0.8754	0.4734	0.6659	0.7198	0.116
28	65.2	0.0	0.0	1.0	0.0	0.0
29	69.87	0.0	0.3405	0.0	1.0	1.0
30	66.07	0.0	1.0	1.0	1.0	0.0
31	72.03	0.5285	0.656	0.2397	0.1479	0.8395
32	68.82	1.0	0.0	0.0	0.3732	1.0
33	62.83	0.0	0.0	0.0	0.0	1.0
34	64.66	0.0	0.0	1.0	1.0	1.0
35	67.75	1.0	1.0	1.0	1.0	1.0
36	79.43	0.6047	0.4385	0.4091	0.9044	0.09717
37	66.8	0.3672	0.07047	0.337	0.03224	0.0527
38	73.37	0.6025	0.2698	0.6621	0.6134	0.4673
39	82.69	1.0	0.6357	0.0	1.0	0.2748
40	74.14	0.3234	0.8398	0.2268	0.4203	0.3872
41	85.73	1.0	0.5833	0.01206	0.9904	0.3604
42	75.48	0.4931	0.1671	0.3198	0.7225	0.2882
43	70.64	0.1547	0.3482	0.3223	0.4066	0.6428
44	66.89	0.2319	0.3344	0.4528	0.08783	0.7829
45	82.58	1.0	0.7641	0.01501	0.979	0.3842
46	83.45	0.998	0.548	0.0004086	0.9881	0.3025
47	83.4	1.0	0.6476	0.01099	1.0	0.3621
48	84.17	0.9913	0.5896	0.04902	1.0	0.3198
49	83.73	0.9794	0.7768	0.01112	0.9502	0.2871
50	84.03	0.9864	0.7574	0.06849	1.0	0.3007
51	84.31	1.0	0.5264	0.0	1.0	0.4464
52	73.48	0.3185	0.7488	0.5395	0.7364	0.7482
53	81.84	0.9313	0.5703	0.0	1.0	0.3985
54	82.67	1.0	0.5861	0.0	0.9837	0.4361
55	85.35	0.9108	0.594	0.04572	0.9676	0.43
56	65.08	0.08285	0.1152	0.6741	0.02296	0.5019
57	84.02	0.9496	0.5849	0.09675	0.9586	0.4755
58	65.47	0.06145	0.08404	0.5739	0.1334	0.9928
59	79.62	0.8617	0.5617	0.1675	0.406	0.3655
60	86.08	0.8043	0.6533	0.07883	0.916	0.416
61	71.08	0.5982	0.6869	0.3947	0.2382	0.7861
62	85.35	0.8613	0.6319	0.05229	0.9338	0.485
63	83.45	0.847	0.5874	0.09521	0.9158	0.4259
64	83.09	0.8734	0.6685	0.08666	0.9809	0.4399
65	83.59	0.9083	0.633	0.0461	0.9128	0.4407
66	67.21	0.2775	0.8315	0.8527	0.3959	0.08389
67	69.58	0.9174	0.3356	0.6946	0.2725	0.3205
68	86.52	0.8742	0.6934	0.0978	0.894	0.3474
69	69.23	0.297	0.848	0.6868	0.9933	0.9283
70	83.08	0.8164	0.7011	0.06874	0.8857	0.3991
71	72.43	0.469	0.9601	0.2361	0.2819	0.3908
72	81.61	0.891	0.6975	0.03869	0.8507	0.3522
73	74.0	0.5354	0.9464	0.02789	0.3534	0.6488
74	70.88	0.3675	0.1983	0.561	0.9438	0.07525
75	81.14	0.8982	0.6471	0.01289	0.938	0.4318
76	85.58	0.9057	0.6741	0.1335	0.9024	0.3723
77	83.83	0.8232	0.7204	0.2572	0.8301	0.6554
78	71.4	0.9517	0.1744	0.5244	0.5347	0.778
79	84.32	0.8329	0.6559	0.1201	0.8918	0.5976
80	83.4	0.8728	0.5645	0.04166	0.9719	0.5724
81	72.62	0.2255	0.1504	0.2482	0.5797	0.3709
82	71.99	0.9694	0.0001661	0.3554	0.494	0.2382
83	78.57	0.9036	0.7302	0.3616	0.9882	0.6229
84	83.47	0.8838	0.6616	0.09444	0.9274	0.3424
85	70.78	0.2932	0.6917	0.2411	0.2711	0.7546
86	82.52	0.741	0.7573	0.2139	0.8413	0.5223
87	71.8	0.6605	0.1678	0.6896	0.9174	0.7587
88	86.05	0.8748	0.7257	0.1316	0.8738	0.3862
89	69.26	0.1561	0.2754	0.82	0.7547	0.4299
90	85.52	0.8634	0.6658	0.13	0.8808	0.4639
91	85.35	0.8772	0.7411	0.1359	0.8664	0.2966
92	80.8	0.831	0.3091	0.283	0.8827	0.2526
93	85.24	0.8673	0.6894	0.1262	0.8603	0.3585
94	71.76	0.348	0.9189	0.3598	0.3521	0.6246
95	84.5	0.8953	0.7286	0.1045	0.8994	0.3504
96	68.86	0.532	0.4544	0.784	0.2509	0.6545
97	81.62	0.88	0.7671	0.1996	0.8653	0.5468
98	72.96	0.8335	0.3828	0.6413	0.3403	0.2382
99	84.74	0.7958	0.7655	0.1715	0.8928	0.3043
100	85.17	0.8691	0.6883	0.1191	0.9011	0.4003
101	84.21	0.8183	0.6219	0.2278	0.8147	0.5809
102	84.43	0.8582	0.788	0.2186	0.8336	0.3736
103	67.61	0.3372	0.4116	0.9822	0.6813	0.05155
104	74.2	0.4219	0.5054	0.3729	0.5243	0.05842
105	70.94	0.4733	0.6679	0.1035	0.03004	0.2836

iter	target	A	B	C	D	E
1	67.02	0.0651	0.8124	0.7497	0.3086	0.2143
2	72.95	0.3379	0.9082	0.4014	0.7404	0.7779
3	65.72	0.09457	0.06615	0.6378	0.1791	0.4731
4	69.08	0.2608	0.08497	0.3713	0.8028	0.8781
5	65.38	0.1084	0.4137	0.6095	0.05669	0.2846
6	72.16	0.4276	1.0	0.3157	0.9537	0.9425
7	70.47	0.3539	1.0	0.2178	0.5244	0.9678
8	73.19	0.3614	0.9631	0.5275	0.9601	0.6467
9	69.08	0.2099	0.9562	0.7198	0.9214	0.9056
10	77.85	0.4579	0.9367	0.3199	0.9015	0.5789
11	80.47	0.5369	0.9307	0.1998	0.9124	0.4749
12	80.91	0.6429	0.9324	0.05015	0.945	0.3284
13	82.67	0.8398	0.8897	0.1928	0.8571	0.4029
14	77.43	1.0	1.0	0.2905	0.7368	0.2649
15	82.07	0.8792	0.7232	0.1182	0.9649	0.505
16	84.2	0.8294	0.8862	0.09002	0.8356	0.5221
17	77.13	0.9594	1.0	0.0	0.8924	0.6147
18	81.73	0.8556	0.8869	0.0639	0.8388	0.5728
19	67.91	0.9136	0.1475	0.9296	0.5193	0.2632
20	76.52	0.6482	0.9165	0.4041	0.7935	0.1456
21	77.51	0.6746	0.8923	0.4183	0.8364	0.1648
22	84.98	0.7509	0.8292	0.07682	0.8009	0.456
23	79.83	0.7528	0.8718	0.09691	0.8557	0.4803
24	84.61	0.8446	0.821	0.08417	0.7561	0.4526
25	84.02	0.7362	0.7866	0.05213	0.7488	0.4249
26	76.12	0.2812	0.2693	0.1092	0.8585	0.3997
27	84.4	0.8318	0.8336	0.1352	0.7872	0.5027
28	84.86	0.8119	0.8817	0.05853	0.7707	0.4575
29	85.21	0.8332	0.8095	0.05654	0.8265	0.4469
30	71.67	0.2311	0.4083	0.3237	0.492	0.05882
31	70.41	0.954	0.696	0.2737	0.1384	0.5637
32	80.81	0.9094	0.8819	0.1074	0.836	0.4642
33	83.76	0.7949	0.8098	0.05788	0.783	0.4829
34	85.49	0.785	0.8155	0.1032	0.7913	0.3863
35	85.38	0.6052	0.5235	0.1254	0.8308	0.6538
36	70.97	0.6986	0.3871	0.4785	0.2144	0.7079
37	84.21	0.6537	0.5533	0.1472	0.8607	0.5455
38	82.48	0.5288	0.5723	0.09881	0.823	0.5834
39	80.69	0.6658	0.4769	0.09406	0.915	0.6498
40	84.78	0.6927	0.5295	0.1773	0.7711	0.6448
41	76.91	0.7111	0.9194	0.3331	0.6995	0.6491
42	83.16	0.5986	0.5348	0.2238	0.8132	0.6374
43	85.09	0.7908	0.676	0.124	0.8594	0.3921
44	73.57	0.4445	0.2263	0.6531	0.6419	0.4605
45	69.17	0.7967	0.6866	0.9581	0.8829	0.8328
46	82.94	0.7946	0.7278	0.01856	0.8876	0.3413
47	83.41	0.6275	0.4984	0.09918	0.7431	0.6248
48	82.42	0.7862	0.6335	0.233	0.8197	0.4694
49	64.66	0.151	0.04546	0.8039	0.1409	0.7411
50	82.74	0.6449	0.5943	0.1254	0.8072	0.6824
51	66.02	0.68	0.3821	0.699	0.07644	0.9617
52	86.29	0.6269	0.7116	0.1432	0.8414	0.3474
53	69.12	0.07694	0.7454	0.2694	0.1914	0.237
54	84.53	0.4784	0.7722	0.1258	0.9379	0.2819
55	84.66	0.5449	0.802	0.1024	0.7952	0.3198
56	84.75	0.6373	0.6578	0.185	0.9503	0.2826
57	68.99	0.8156	0.8434	0.0135	0.3931	0.6865
58	66.58	0.1542	0.2361	0.4916	0.19	0.7341
59	82.95	0.5716	0.658	0.0621	0.8862	0.3157

60	72.03	0.3452	0.8698	0.1896	0.7661	0.9465
61	66.41	0.241	0.6871	0.9927	0.4714	0.9133
62	84.11	0.556	0.7467	0.2368	0.8738	0.3335
63	85.03	0.8851	0.7102	0.1463	0.9203	0.5214
64	82.71	0.962	0.593	0.1843	0.8805	0.5532
65	76.34	0.9256	0.7548	0.5043	0.4022	0.4134
66	84.09	0.6372	0.7712	0.1437	0.8794	0.2487
67	76.09	0.3781	0.9696	0.09194	0.9306	0.2828
68	81.67	0.9124	0.7252	0.3015	0.9385	0.5394
69	80.26	0.4662	0.6404	0.2285	1.0	0.1701
70	82.69	0.5875	0.7475	0.1344	0.9049	0.3618
71	83.39	0.4839	0.7853	0.09529	0.9628	0.2987
72	84.53	0.7599	0.8133	0.1324	0.7687	0.4067
73	66.64	0.4337	0.1128	0.7631	0.1971	0.2853
74	83.84	0.6305	0.6409	0.1783	0.8104	0.3007
75	86.43	0.6239	0.6825	0.1675	0.8015	0.2947
76	82.42	0.4701	0.752	0.214	0.6843	0.3155
77	72.44	0.9631	0.6141	0.6156	0.3545	0.4719
78	82.54	0.83	0.6654	0.2626	1.0	0.2368
79	77.28	0.6738	0.4118	0.1761	0.7466	0.8095
80	73.32	0.3252	0.1426	0.2134	0.7401	0.3447
81	84.22	0.6408	0.7332	0.001911	0.7904	0.1763
82	72.17	0.4775	0.04031	0.0196	0.2778	0.469
83	73.57	0.9418	0.07991	0.3573	0.9119	0.7214
84	86.08	0.6294	0.541	0.2187	0.8359	0.6128
85	82.43	0.7019	0.5455	0.2356	0.8764	0.612
86	65.83	0.1881	0.09081	0.7714	0.1559	0.5161
87	70.29	0.7824	0.07968	0.7231	0.773	0.1294
88	77.58	0.4993	0.7023	0.009668	0.792	0.07027
89	84.53	0.6315	0.7347	0.1184	0.9129	0.3691
90	83.23	0.7913	0.6921	0.1181	0.8617	0.3741
91	71.31	0.6197	0.4162	0.5782	0.3633	0.8594
92	71.34	0.3686	0.2240	0.1972	0.4807	0.5209
93	79.56	0.5957	0.7235	0.03062	0.5858	0.2452
94	80.11	0.5265	0.54	0.1048	0.7932	0.6104
95	84.61	0.6593	0.5232	0.2284	0.838	0.6012
96	84.4	0.8315	0.5654	0.03341	0.855	0.4107
97	71.51	0.08371	0.6647	0.5289	0.5758	0.3213
98	82.4	0.6883	0.8239	0.0	0.8005	0.3014
99	69.06	0.2804	0.4737	0.3962	0.1014	0.3111
100	82.88	0.7969	0.6511	0.1107	0.8611	0.4645
101	83.24	0.6014	0.7336	0.1503	0.7719	0.2859
102	83.67	0.8679	0.5912	0.1669	0.8852	0.3845
103	84.99	0.7289	0.6078	0.0	0.8223	0.3089
104	73.44	0.291	0.179	0.05806	0.9068	0.5405
105	82.63	0.6861	0.6802	0.0	0.9484	0.2035

iter	target	A	B	C	D	E
1	70.92	0.1457	0.7865	0.5337	0.596	0.03081
2	71.31	0.1532	0.5814	0.135	0.6082	0.8789
3	68.43	0.1709	0.9457	0.8219	0.7227	0.4041
4	71.34	0.06685	0.9907	0.1548	0.8531	0.327
5	71.88	0.5482	0.7297	0.2053	0.09352	0.2253
6	68.38	0.615	0.194	0.9473	0.684	0.08124
7	68.23	0.00859	0.1321	0.1377	0.4147	0.7567
8	68.31	0.3132	0.1137	0.461	0.1362	0.4245
9	70.89	0.8492	0.4542	0.2738	0.2547	0.8849
10	80.7	0.6779	0.8652	0.2934	0.9671	0.3773
11	69.73	0.5024	0.2812	0.4037	0.3158	0.0572
12	79.28	0.5968	0.8514	0.2855	0.9696	0.2825
13	78.25	0.858	0.9112	0.2171	1.0	0.4225
14	82.67	0.6621	0.7879	0.334	1.0	0.49
15	78.42	0.6826	0.7776	0.4349	1.0	0.6202
16	84.27	0.6439	0.6968	0.2364	1.0	0.4706
17	83.1	0.6244	0.7095	0.2033	0.9767	0.4681
18	80.4	0.7976	0.4355	0.2821	0.9991	0.2397
19	83.71	0.6964	0.6268	0.287	1.0	0.4367
20	69.68	0.5098	0.3351	0.4324	0.09673	0.5314
21	83.69	0.7301	0.5206	0.1279	1.0	0.595
22	78.44	0.9726	0.1982	0.0	1.0	0.5429
23	66.28	0.06761	0.5825	0.3469	0.1719	0.8822
24	79.99	0.373	0.7101	0.2536	0.9619	0.4626
25	84.56	0.5898	0.5484	0.2167	1.0	0.5515
26	80.05	0.504	0.3602	0.0	1.0	0.6757
27	74.41	0.5119	0.0	0.0	1.0	0.0
28	73.77	0.7773	0.5387	0.0	1.0	1.0
29	85.82	0.639	0.5934	0.2144	1.0	0.5261
30	65.53	1.0	1.0	1.0	0.0	1.0
31	68.47	1.0	0.0	0.0	5.551e-17	0.0
32	84.35	0.6641	0.5578	0.1557	1.0	0.4828
33	70.84	1.0	1.0	0.6967	1.0	0.0
34	72.08	1.0	0.0	0.4675	1.0	0.4897
35	81.3	0.6625	0.7346	0.4565	0.9735	0.5003
36	79.41	1.0	0.4401	0.0	1.0	0.0
37	86.6	0.659	0.6141	0.2499	1.0	0.5543
38	86.35	0.6343	0.6043	0.3211	1.0	0.5628
39	64.64	0.0	0.0	1.0	1.0	1.0
40	73.76	0.8504	0.3795	0.6419	0.9288	0.4078
41	78.17	1.0	1.0	0.0	0.5424	0.0
42	84.01	0.7284	0.5447	0.1324	0.9861	0.6082
43	69.71	0.2364	0.8989	0.2952	0.6553	0.9442
44	68.38	0.2551	0.2185	0.5913	0.3159	0.2855
45	84.95	0.6636	0.6065	0.2907	0.929	0.5757
46	65.89	1.0	1.0	1.0	0.0	0.0
47	65.11	0.07411	0.03299	0.4992	0.05035	0.7081
48	81.01	1.0	0.5221	0.0	0.7013	0.3149
49	64.37	1.0	0.0	1.0	0.0	1.0
50	66.09	0.0	0.0	0.0	1.0	1.0
51	65.44	0.2395	0.633	0.8904	0.09965	0.2156
52	73.34	1.0	1.0	0.0	0.3578	0.505
53	73.95	1.0	0.1283	0.0	0.7363	0.1387
54	75.62	0.9945	0.9851	0.0008635	0.5442	0.01861
55	78.7	0.4851	0.5982	0.4912	1.0	0.5072
56	72.97	0.9769	0.9479	0.6148	0.521	0.5516
57	81.21	1.0	0.5278	0.0	1.0	0.3807
58	85.27	0.7855	0.5799	0.2929	1.0	0.5857
59	82.42	0.5566	0.6696	0.0	1.0	0.6513
60	75.16	1.0	1.0	0.0	1.0	0.0
61	80.37	0.5878	0.2881	0.4703	0.8931	0.469
62	69.72	0.5181	0.6537	0.334	0.05112	0.6584
63	78.24	0.7066	0.3106	0.0009428	0.745	0.4312
64	67.8	0.5942	0.006378	0.09804	0.0544	0.003447
65	71.09	0.0	0.3673	0.0	1.0	0.0
66	70.04	1.0	1.0	0.0	0.05299	0.0
67	67.98	0.1416	0.4863	0.7292	0.4158	0.8423
68	82.29	0.9667	0.5157	0.06297	0.7228	0.5861
69	74.43	0.4253	0.3268	0.4457	0.9772	0.7704
70	69.06	0.0	1.0	0.0	0.0	0.0
71	77.88	1.0	0.709	0.0	0.9011	0.6424
72	67.65	0.1792	0.9438	0.2614	0.1325	0.6459
73	68.25	0.214	0.254	0.9982	0.7864	0.2607
74	84.61	0.6313	0.6551	0.2051	0.9973	0.6271
75	80.08	1.0	0.424	0.189	0.7955	0.411
76	70.62	1.0	0.0	0.0	0.6557	1.0
77	67.97	1.0	1.0	0.0	0.0	1.0
78	80.15	0.732	0.3125	0.002277	0.6999	0.4251
79	75.57	0.9288	0.3666	0.0	0.5272	0.4042
80	82.93	0.8257	0.5608	0.0	0.7834	0.4802
81	82.43	0.7025	0.3892	0.2631	0.6571	0.2812
82	84.54	0.7493	0.5056	0.1934	0.6456	0.4731
83	85.98	0.7044	0.3595	0.3178	0.6474	0.508
84	82.97	0.6871	0.4603	0.3865	0.6566	0.4868
85	69.33	0.2124	0.1616	0.7805	0.7115	0.4876
86	70.14	0.172	0.5859	0.3122	0.4406	0.8497
87	84.16	0.6576	0.3417	0.2289	0.5906	0.4826
88	86.31	0.765	0.3644	0.2521	0.6922	0.4926
89	66.0	0.7394	0.00716	0.8166	0.2457	0.8012
90	83.47	0.7453	0.3792	0.2625	0.685	0.5979
91	82.64	0.7569	0.3025	0.3075	0.6516	0.4439
92	84.14	0.6975	0.3981	0.2615	0.7096	0.4725
93	85.55	0.7828	0.4041	0.2462	0.6337	0.4875
94	83.06	0.5698	0.7123	0.05281	0.8963	0.2246
95	82.7	0.6821	0.5218	0.06476	0.7783	0.09331
96	78.52	0.5981	0.6284	0.0	1.0	0.01126
97	85.75	0.7766	0.491	0.1032	0.6818	0.3077
98	70.71	0.5963	0.3524	0.4101	0.4088	0.9977
99	80.96	0.6326	0.5179	0.0608	0.5089	0.2619
100	84.47	0.5871	0.6335	0.332	0.9958	0.5315
101	84.56	0.8178	0.4483	0.1494	0.7452	0.3777

102	79.21	0.3944	0.6697	0.0	1.0	0.3713
103	74.67	0.9355	0.1923	0.4132	0.5964	0.2175
104	86.36	0.7791	0.5831	0.1653	0.75	0.2266
105	81.35	0.7848	0.5972	0.07542	0.8289	0.2402

iter	target	A	B	C	D	E
1	68.34	0.03389	0.04659	0.6123	0.9388	0.3068
2	67.37	0.07117	0.9298	0.8506	0.3809	0.4504
3	76.21	0.6093	0.4544	0.6342	0.788	0.3225
4	83.09	0.7272	0.6771	0.3031	0.7002	0.6685
5	65.39	0.3523	0.9892	0.8199	0.8316	0.6371
6	82.29	0.7572	0.6252	0.3729	0.681	0.6673
7	83.45	0.9503	0.7245	0.05247	0.868	0.7444
8	72.64	0.6674	0.8408	0.1775	0.9921	1.0
9	69.29	0.3221	0.03175	0.1524	0.1126	0.4987
10	83.6	0.9298	0.693	0.1222	0.7137	0.552
11	79.38	0.879	0.5458	0.04788	0.7104	0.7232
12	67.75	0.2675	0.01988	0.544	0.04908	0.3407
13	81.09	0.958	0.838	0.2336	0.8405	0.5804
14	77.86	0.9961	0.6758	0.09008	0.8377	0.7453
15	80.16	0.8623	0.7843	0.008785	0.8585	0.6738
16	81.36	0.8379	0.693	0.2235	0.7055	0.5798
17	80.97	0.9691	0.7919	0.1052	0.6825	0.5068
18	68.62	0.1077	0.4527	0.8023	0.8822	0.2791
19	77.85	0.9521	0.78	0.02329	0.9405	0.7809
20	85.03	0.8934	0.7373	0.1048	0.7953	0.673
21	74.68	0.3755	0.5714	0.4202	0.4068	0.5405
22	82.33	0.8795	0.6846	0.1093	0.8476	0.6115
23	79.27	0.3123	0.6839	0.3183	0.8514	0.3851
24	77.83	0.8603	0.7372	0.08812	0.7891	0.7607
25	78.95	0.9418	0.7604	0.08779	0.7975	0.6345
26	81.22	0.8639	0.711	0.1376	0.7717	0.6458
27	80.24	0.8949	0.7242	0.09559	0.8476	0.6879
28	81.66	0.7866	0.6598	0.3098	0.6898	0.6282
29	83.82	0.9121	0.7261	0.1723	0.7394	0.521
30	78.18	0.2872	0.673	0.312	0.8913	0.3611
31	78.58	0.4383	0.6348	0.2451	0.4134	0.234
32	70.04	0.6658	0.001072	0.3824	0.3659	0.1337
33	83.54	0.722	0.6890	0.3706	0.7335	0.6441
34	79.79	0.8881	0.77	0.1119	0.7618	0.6878
35	83.29	0.8902	0.6722	0.1148	0.7735	0.5373
36	67.92	0.9486	0.0007625	0.6851	0.6194	0.9839
37	69.76	0.7053	0.327	0.7406	0.3321	0.6633
38	73.23	0.9344	0.8859	0.7174	0.6039	0.3006
39	82.60	0.8001	0.6918	0.1253	0.6914	0.4995
40	69.89	0.32	0.2667	0.0002091	0.2582	0.1962
41	83.63	0.6887	0.6358	0.3375	0.7159	0.6204
42	83.59	0.6932	0.6875	0.3583	0.6619	0.629
43	67.33	0.998	0.04005	0.8783	0.849	0.9272
44	65.77	0.3434	0.2947	0.5853	0.05406	0.7882
45	83.34	0.6847	0.6312	0.3419	0.7458	0.5746
46	69.39	0.8426	0.8231	0.992	0.7816	0.5498
47	69.41	0.9914	0.354	0.8028	0.8046	0.078
48	75.45	0.6778	0.9358	0.4699	0.9148	0.2067
49	67.12	0.2359	0.4651	0.8259	0.1237	0.4171
50	72.06	0.8346	0.5935	0.6024	0.9529	0.5235
51	73.0	0.3879	0.4316	0.6767	0.8995	0.1927
52	81.8	0.7014	0.711	0.3107	0.72	0.5754
53	83.52	0.933	0.6323	0.1013	0.7205	0.5206
54	83.78	0.8921	0.7041	0.06506	0.782	0.6646
55	68.89	0.4271	0.1766	0.4322	0.02221	0.0103
56	73.93	0.5966	0.1172	0.3558	0.4367	0.5411
57	68.36	0.3409	0.266	0.8198	0.4218	0.2661
58	74.67	0.2753	0.413	0.3485	0.921	0.8055
59	72.65	0.286	0.7408	0.1096	0.7114	0.8884
60	80.23	0.9626	0.6945	0.185	0.6528	0.5106
61	84.88	0.889	0.6513	0.2231	0.8103	0.5278
62	84.24	0.8385	0.6241	0.2236	0.7698	0.5003
63	72.6	0.1079	0.0316	0.01505	0.8326	0.5624
64	85.51	0.7796	0.7173	0.2386	0.7083	0.6017
65	85.14	0.7712	0.709	0.2365	0.7777	0.5667
66	72.06	0.5805	0.7317	0.42	0.2779	0.05484
67	81.38	0.8062	0.6565	0.3219	0.6884	0.6433
68	84.71	0.7696	0.7635	0.2478	0.7354	0.597
69	84.39	0.7368	0.7179	0.2052	0.7183	0.5812
70	85.31	0.8304	0.6993	0.2191	0.8368	0.494
71	82.77	0.8073	0.6513	0.2705	0.8482	0.5319
72	82.69	0.8961	0.6612	0.1945	0.8164	0.4513
73	85.33	0.8438	0.7373	0.2584	0.8007	0.5145
74	69.89	0.8494	0.9762	0.0518	0.1496	0.6957
75	66.23	0.2288	0.05281	0.1717	0.008132	0.6196
76	75.26	0.7249	0.6201	0.248	0.5076	0.002373
77	75.32	0.844	0.2512	0.3939	0.4208	0.7024
78	80.55	0.797	0.7237	0.2184	0.7745	0.496
79	83.01	0.7582	0.712	0.2445	0.751	0.6172
80	70.33	0.6159	0.7042	0.9367	0.8458	0.8378
81	82.57	0.86	0.7305	0.2342	0.8475	0.538
82	84.49	0.8277	0.6335	0.1758	0.8389	0.517
83	67.58	0.9975	0.2085	0.9246	0.2749	0.5127
84	82.53	0.9145	0.7882	0.3101	0.755	0.5491
85	72.84	0.2404	0.1419	0.127	0.5604	0.3604
86	81.53	0.6332	0.637	0.4249	0.726	0.6525
87	81.97	0.7568	0.7514	0.2457	0.6534	0.6069
88	87.05	0.8686	0.6961	0.2883	0.8018	0.4928
89	85.32	0.8702	0.7344	0.3254	0.8283	0.4587
90	76.29	0.4515	0.4143	0.007778	0.6849	0.164
91	69.6	0.6476	0.3234	0.8941	0.7893	0.0009895
92	83.59	0.8395	0.7124	0.3388	0.7798	0.5186
93	83.48	0.9239	0.7141	0.2883	0.7988	0.4738
94	74.29	0.5973	0.52	0.5792	0.4256	0.1002
95	84.01	0.9496	0.5894	0.05392	0.6949	0.5009
96	81.94	0.8883	0.5118	0.1463	0.7243	0.5321
97	76.38	0.7177	0.7381	0.4084	0.6571	0.7468
98	84.23	0.8519	0.6841	0.2796	0.8313	0.4669
99	82.78	0.8888	0.5581	0.02746	0.8094	0.4969
100	81.06	0.8392	0.5845	0.002383	0.6817	0.5182
101	81.93	0.8864	0.8503	0.3368	0.8438	0.4664
102	81.62	1.0	0.5649	0.1039	0.7841	0.4915
103	67.66	0.2756	0.3196	0.9166	0.5118	0.7861
104	83.94	0.9562	0.6738	0.03365	0.7186	0.4278
105	84.21	0.8034	0.669	0.06611	0.9051	0.4776

iter	target	A	B	C	D	E
1	75.5	0.499	0.2114	0.2524	0.6525	0.122
2	73.64	0.7106	0.1135	0.3848	0.3641	0.4339
3	66.82	0.6303	0.9647	0.9944	0.2619	0.0787
4	74.03	0.5095	0.6313	0.6659	0.9938	0.728
5	72.3	0.4803	0.5957	0.6085	0.4622	0.7086
6	74.5	0.4225	0.106	0.1135	1.0	0.3135
7	71.38	0.3321	0.0	0.0	0.5094	0.0
8	68.09	0.7904	0.8581	0.7399	0.1393	0.5692
9	68.25	0.5585	0.9716	0.9153	0.721	0.7362
10	79.45	0.535	0.3067	0.3929	0.8501	0.3241
11	76.76	0.6183	0.2933	0.5239	0.9697	0.2935
12	72.17	0.7966	0.3362	0.2819	0.1596	0.4825
13	83.03	0.4688	0.3951	0.3435	0.8258	0.4141
14	84.35	0.4038	0.4815	0.2781	0.8281	0.4635
15	67.37	0.3648	0.555	0.7301	0.1135	0.3711
16	81.35	0.5281	0.5314	0.1705	0.8262	0.5088
17	70.65	0.3616	0.8288	0.498	0.2154	0.06192
18	70.84	0.5605	0.692	0.9803	0.5916	0.5656
19	78.66	0.256	0.4713	0.3205	0.839	0.4432
20	84.39	0.4643	0.5308	0.3071	0.8362	0.4235
21	82.9	0.4583	0.495	0.3176	0.8316	0.5045
22	84.79	0.4384	0.4913	0.2407	0.8187	0.3856
23	86.74	0.4407	0.5005	0.2494	0.9216	0.3988
24	72.95	0.5084	0.8551	0.2357	0.2375	0.2161
25	81.61	0.4445	0.5716	0.2189	1.0	0.3498
26	83.47	0.4976	0.3575	0.1913	0.9584	0.3822
27	69.56	0.00093	0.745	0.2376	0.6702	0.8708
28	64.5	0.0	0.0	1.0	0.0	1.0
29	66.11	0.9045	0.04518	0.2272	0.05334	0.903
30	80.42	0.4105	0.5129	0.276	0.9928	0.4622
31	80.49	0.7621	0.6256	0.008795	0.5775	0.5212
32	70.96	0.2697	0.5118	0.8165	0.8273	0.5711
33	72.78	0.8658	0.3709	0.1276	0.2894	0.7787
34	75.68	0.3722	0.9191	0.06089	0.673	0.8246
35	73.44	0.1678	0.4261	0.107	0.5863	0.4643

36	68.53	0.4833	0.9968	0.6764	0.919	0.9616
37	83.31	0.4595	0.4604	0.2504	0.9005	0.3792
38	84.71	0.4185	0.5376	0.2233	0.8959	0.4136
39	73.03	0.5418	0.1708	0.08723	0.877	0.9386
40	81.5	0.414	0.5104	0.2855	0.9125	0.3906
41	76.21	0.3913	0.9983	0.2517	0.8745	0.5404
42	76.31	0.368	0.539	0.3401	0.5343	0.2588
43	84.24	0.4634	0.5019	0.2243	0.9284	0.4298
44	81.66	0.4152	0.4885	0.2163	0.9419	0.3926
45	78.31	0.5193	0.286	0.392	0.8577	0.3549
46	81.38	0.457	0.5121	0.2534	0.8811	0.4207
47	84.57	0.4432	0.4874	0.3837	0.7866	0.4223
48	83.32	0.4168	0.5399	0.2581	0.7896	0.4255
49	85.23	0.4163	0.431	0.2514	0.795	0.4149
50	69.02	0.5607	0.388	0.5578	0.1087	0.6315
51	79.85	0.4543	0.4632	0.2302	0.7763	0.4501
52	72.37	0.5539	0.3765	0.1244	0.2478	0.1608
53	66.89	0.1847	0.6638	0.6755	0.1812	0.7946
54	84.27	0.4501	0.5222	0.2454	0.9437	0.3912
55	83.56	0.3952	0.4596	0.2787	0.8138	0.3888
56	80.63	0.5665	0.6574	0.3389	0.9375	0.6785
57	84.47	0.4201	0.4017	0.2698	0.8556	0.4401
58	68.12	0.03811	0.708	0.6071	0.6988	0.9242
59	81.7	0.4531	0.5406	0.3595	0.8002	0.4413
60	73.5	0.444	0.6732	0.6286	0.3562	0.4112
61	69.04	0.6699	0.4422	0.928	0.754	0.02905
62	70.74	0.1464	0.9716	0.5572	0.4745	0.3574
63	66.84	0.03032	0.2002	0.8175	0.6719	0.8403
64	76.34	0.1713	0.4844	0.3026	0.7401	0.3814
65	81.2	0.4504	0.3878	0.2567	0.814	0.3756
66	82.49	0.4416	0.4752	0.263	0.9359	0.4225
67	81.38	0.4058	0.4131	0.3054	0.8031	0.4575
68	83.41	0.3941	0.4481	0.2183	0.8374	0.4187
69	80.89	0.4838	0.3437	0.2204	0.9036	0.4501
70	80.27	0.7545	0.6838	0.0233	0.5934	0.4832
71	79.83	0.4702	0.5276	0.2969	0.7916	0.3611
72	79.05	0.5629	0.5272	0.523	0.9932	0.6037
73	83.12	0.4568	0.4668	0.324	0.8344	0.4326
74	82.15	0.4523	0.5088	0.2268	0.7439	0.5733
75	66.89	0.7779	0.7586	0.9124	0.1385	0.05299
76	81.46	0.4373	0.5081	0.2337	0.9118	0.3906
77	66.64	0.5103	0.1003	0.8836	0.9912	0.9335
78	80.75	0.4176	0.5365	0.2376	0.9274	0.4395
79	84.93	0.3979	0.5414	0.218	0.854	0.427
80	84.14	0.5573	0.5714	0.03257	0.9635	0.3318
81	71.39	0.8477	0.2936	0.2369	0.3829	0.8385
82	82.02	0.4313	0.4545	0.265	0.8252	0.418
83	82.77	0.4203	0.5204	0.3073	0.8176	0.4495
84	80.08	0.3817	0.3966	0.245	0.8128	0.4198
85	69.83	0.5165	0.7776	0.7278	0.8117	0.9059
86	68.69	0.1683	0.2237	0.8097	0.736	0.4507
87	84.98	0.4039	0.4776	0.2363	0.783	0.3928
88	81.96	0.4408	0.4975	0.2509	0.9321	0.398
89	82.24	0.4284	0.492	0.2482	0.9144	0.4138
90	75.51	0.4642	0.6222	0.1966	0.8706	0.9675
91	78.6	0.408	0.4982	0.2501	0.8107	0.4142
92	67.21	0.3459	0.09816	0.8808	0.6809	0.9139
93	71.62	0.4384	0.259	0.5176	0.442	0.02697
94	68.66	0.3295	0.7862	0.4148	0.05341	0.2171
95	83.91	0.4215	0.4536	0.2408	0.7968	0.3885
96	86.5	0.4491	0.544	0.2128	0.7929	0.363
97	72.44	0.9301	0.2543	0.6263	0.7485	0.7075
98	85.68	0.4416	0.5206	0.2255	0.7942	0.3752
99	84.32	0.4414	0.5498	0.2219	0.8152	0.3853
100	76.18	0.6991	0.5628	0.4293	0.5563	0.7358
101	84.31	0.4431	0.5231	0.2094	0.8107	0.3542
102	70.43	0.4357	0.7191	0.7989	0.2814	0.5458
103	72.32	0.6752	0.5135	0.5781	0.3669	0.07575
104	79.83	0.4337	0.4106	0.2979	0.8284	0.4136
105	65.65	0.008919	0.1087	0.6837	0.06049	0.2812

iter	target	A	B	C	D	E
1	68.25	0.2146	0.04636	0.6485	0.5165	0.3694
2	68.62	0.2702	0.7335	0.2781	0.1102	0.4249
3	71.59	0.07945	0.8041	0.4293	0.8087	0.6127
4	66.51	0.5579	0.3499	0.9879	0.1593	0.09351
5	74.51	0.3738	0.6452	0.563	0.9435	0.2174
6	75.73	0.343	0.794	0.5083	1.0	0.2439
7	72.61	0.4398	1.0	0.4232	1.0	0.04629
8	72.84	0.3594	0.8686	0.6635	1.0	0.3538
9	74.54	0.2795	0.7368	0.3779	1.0	0.2283
10	79.41	0.4788	0.7649	0.4408	1.0	0.2719
11	82.33	0.5722	0.7513	0.395	1.0	0.3129
12	81.57	0.682	0.7329	0.3423	1.0	0.365
13	82.55	0.6599	0.7176	0.4001	1.0	0.2549
14	80.8	0.6767	0.7861	0.4559	1.0	0.3142
15	74.27	0.2646	0.7426	0.4999	0.9895	0.2453
16	77.56	0.349	0.7938	0.5096	0.9524	0.1996
17	65.08	0.4609	0.05991	0.9931	0.3101	0.3035
18	84.49	0.6093	0.7384	0.3344	0.8956	0.2589
19	81.78	0.6529	0.6772	0.1961	0.835	0.1644
20	84.13	0.6113	0.767	0.3774	0.7611	0.2578
21	83.19	0.5658	0.8182	0.2876	0.7673	0.3383
22	68.29	0.3589	0.2849	0.9164	0.4829	0.3988
23	76.27	0.5695	0.6876	0.206	0.9889	0.878
24	70.14	0.1924	0.09116	0.3906	0.6992	0.09841
25	69.99	0.8637	0.03804	0.2239	0.1585	0.5044
26	75.32	0.9061	0.9412	0.3073	0.5958	0.1756
27	83.34	0.5648	0.6645	0.3446	0.8296	0.2906
28	71.71	1.0	0.0	0.0	1.0	1.0
29	83.34	0.5983	0.7554	0.3169	0.8338	0.2471
30	79.21	0.3728	0.7405	0.1371	0.4868	0.3703
31	80.15	0.6271	0.7749	0.4303	0.7742	0.3509
32	77.73	0.3499	0.739	0.1219	0.5238	0.3776
33	79.04	0.553	0.7769	0.3291	0.5905	0.2293
34	82.5	0.6698	0.6823	0.4064	0.8192	0.1785
35	79.5	0.413	0.739	0.1082	0.4949	0.3797
36	82.61	0.6223	0.6274	0.2301	0.9591	0.2781
37	85.41	0.7345	0.6873	0.1112	0.7648	0.3746
38	69.6	0.3639	0.1826	0.4949	0.9775	0.8729
39	75.01	0.3279	0.7892	0.5092	0.9279	0.1758
40	83.95	0.6941	0.6837	0.2091	0.7596	0.3477
41	70.15	0.5686	0.5447	0.5182	0.273	0.8281
42	73.81	0.1774	0.4768	0.2289	0.8851	0.6457
43	85.76	0.688	0.7637	0.08192	0.8277	0.3801
44	70.14	0.2476	0.5959	0.8042	0.5653	0.4904
45	69.61	0.727	0.4998	0.8604	0.5016	0.05804
46	68.62	0.283	0.961	0.458	0.2132	0.08918
47	86.2	0.7772	0.7157	0.05304	0.8613	0.331
48	83.94	0.765	0.7671	0.0	0.771	0.3244
49	71.91	0.2195	0.6236	0.2491	0.3325	0.1124
50	83.71	0.7253	0.6578	0.02171	0.8754	0.4047
51	65.09	0.1828	0.4137	0.9834	0.3661	0.9764
52	83.42	0.7961	0.7691	0.1194	0.8515	0.3755
53	84.84	0.7168	0.7012	0.07353	0.8411	0.2993
54	82.45	0.8852	0.658	0.003361	0.9081	0.2625
55	69.82	0.8865	0.9063	0.4107	0.03029	0.1834
56	84.41	0.7426	0.7962	0.0	0.9609	0.3221
57	81.27	0.6507	0.7755	0.09443	0.7085	0.4838
58	83.38	0.851	0.5466	0.1854	0.7011	0.3198
59	83.53	0.5724	0.8179	0.1626	0.9809	0.3717
60	81.82	0.9107	0.3995	0.1723	0.7734	0.0407
61	75.69	1.0	0.4592	0.0	0.5628	0.06555
62	75.54	0.3244	0.9854	0.3104	0.8157	0.3187
63	82.54	0.9094	0.7261	0.1649	0.7682	0.6366
64	81.2	0.976	0.6	0.02183	0.723	0.5388
65	80.13	0.9095	0.421	0.2577	0.8636	0.285
66	75.23	1.0	0.823	0.03273	0.77	0.8504
67	65.54	0.05597	0.1752	0.9747	0.2858	0.4182
68	85.09	0.8302	0.558	0.1943	0.6947	0.5269
69	78.62	0.9926	0.5577	0.2807	0.6799	0.5956
70	83.92	0.7413	0.5382	0.1263	0.6735	0.4684
71	80.61	0.5935	0.6088	0.5297	0.8688	0.456
72	67.08	0.006628	0.7261	0.4	0.2742	0.7309
73	79.72	0.6013	0.9326	0.0	0.8987	0.3783
74	71.55	0.7982	0.8363	0.3865	0.2597	0.000347
75	84.53	0.813	0.6016	0.147	0.7772	0.4826
76	75.84	0.93	0.1711	0.1426	1.0	0.0
77	82.98	0.7302	0.4682	0.2581	0.8045	0.5526

78	69.11	0.2911	0.06337	0.3274	0.4739	0.06798
79	84.14	0.757	0.499	0.2921	0.6704	0.4014
80	81.62	0.7732	0.5898	0.1718	0.69	0.6738
81	82.73	0.9143	0.7644	0.1629	0.7835	0.6333
82	85.69	0.8215	0.6346	0.1802	0.627	0.4413
83	79.01	0.8697	0.6248	0.198	0.9882	0.0
84	76.16	0.4414	0.5977	0.3286	0.3583	0.4424
85	65.59	1.0	1.0	1.0	1.0	1.0
86	66.03	0.259	0.273	0.6524	0.86641	0.5343
87	71.88	0.209	0.3291	0.1311	0.5765	0.001334
88	66.03	1.0	0.0	1.0	1.0	1.0
89	66.62	1.0	1.0	0.0	0.0	1.0
90	71.28	0.5062	0.2547	0.1025	0.46	0.8988
91	82.76	0.8106	0.5485	0.1994	0.5162	0.449
92	84.99	0.8431	0.6657	0.09493	0.6814	0.4586
93	68.35	1.0	1.0	1.0	1.0	0.0
94	73.1	0.384	0.1647	0.05508	0.5747	0.6115
95	63.93	1.0	0.0	1.0	0.0	1.0
96	67.03	1.0	0.0	1.0	1.0	0.0
97	66.14	1.0	1.0	1.0	0.0	1.0
98	72.52	0.4714	0.6228	0.4816	0.3244	0.6985
99	64.40	0.0	1.0	1.0	1.0	1.0
100	68.66	0.7533	0.8924	0.7745	0.6588	0.9591
101	75.96	1.0	0.4278	0.3904	0.6865	0.0
102	78.46	1.0	0.8266	0.0	1.0	0.5574
103	69.51	0.9946	0.3877	0.2523	0.03735	0.2379
104	82.58	0.8326	0.6379	0.01518	0.8954	0.2765
105	77.83	1.0	1.0	0.0	1.0	0.0